

CHAPTER – 1

ACCIDENTS IN INDIA

Introduction

This is the ninth edition of the report after the comprehensive revision of the proforma for data collection of 'Accidental Deaths & Suicides in India' was done in 2014 in consultation with States/UTs Police and others stakeholders.

An accident is an unplanned event that results in injury, deaths, damage to property or some other losses. For statistical purpose, Accidental Deaths have been classified as deaths that could be due to forces/factors of the nature which have been termed as

'Accidental Deaths due to Forces of Nature' or could be due to deliberate or negligent conduct of human beings, which is termed as 'Other causes of Accidental Deaths'.

The rate of accidental deaths (per lakh of population) has increased in 2022 as compared to 2021. The total number of accidental deaths in 2022 is 4,30,504. [LIST-1.1]. The incidence and rate of accidental deaths during 2012 – 2022 is presented in Table-1.1.

LIST-1.1
Percentage Change in Population, Number and Rate of Accidental Deaths over the Corresponding Previous Year during 2018 to 2022

Year	Mid-Year Projected Population (In Lakh ⁺)	Percentage Change in Population over the Previous Year	Total Number of Accidental Deaths	Percentage Change in Accidental Deaths over the Previous Year	Rate of Accidental Deaths	Percentage Change in Rate of Accidental Deaths over the Previous Year
(1)	(2)	(3)	(4)	(5)	(6)	(7)
2018	13233.8 [#]	1.1	411824	3.8	31.1	2.6
2019	13376.1 [#]	1.1	421104	2.3	31.5	1.3
2020	13533.9 ^{\$}	1.2	374397	-11.1	27.7	-12.1
2021	13671.8 ^{\$}	1.0	397530	6.2	29.1	5.1
2022	13797.5 ^{\$}	0.9	430504	8.3	31.2	7.2

Source: Report of the Technical Group on Population Projections (November, 2019), National Commission on Population, Ministry of Health & Family Welfare.

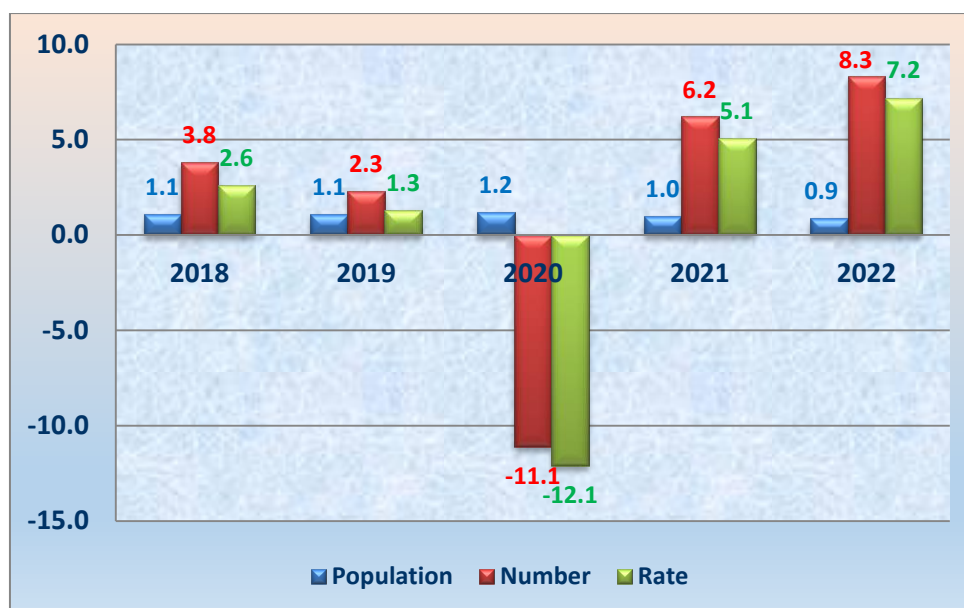
\$ Source: Report of the Technical Group on Population Projections (July, 2020), National Commission on Population, Ministry of Health & Family Welfare

+ One Lakh = 0.1 Million = 100,000

As per data provided by States/UTs

FIGURE-1.1

Percentage Change in Population, Number and Rate of Accidental Deaths during 2018 – 2022 (Over Corresponding Previous Year)



- As per data provided by States/UTs.

Out of 4,30,504 accidental deaths, 8,060 (1.9%) deaths were due to forces of nature and 4,22,444 (98.1%) deaths were due to other causes.

There was an increase of 13.1% (from 7,126 deaths to 8,060 deaths) in deaths due to forces of nature and an increase of 8.2% (from 3,90,404 deaths to 4,22,444 deaths) in deaths due to other causes during 2022 as compared to 2021 **LIST-1.2 and Table 1.3.**

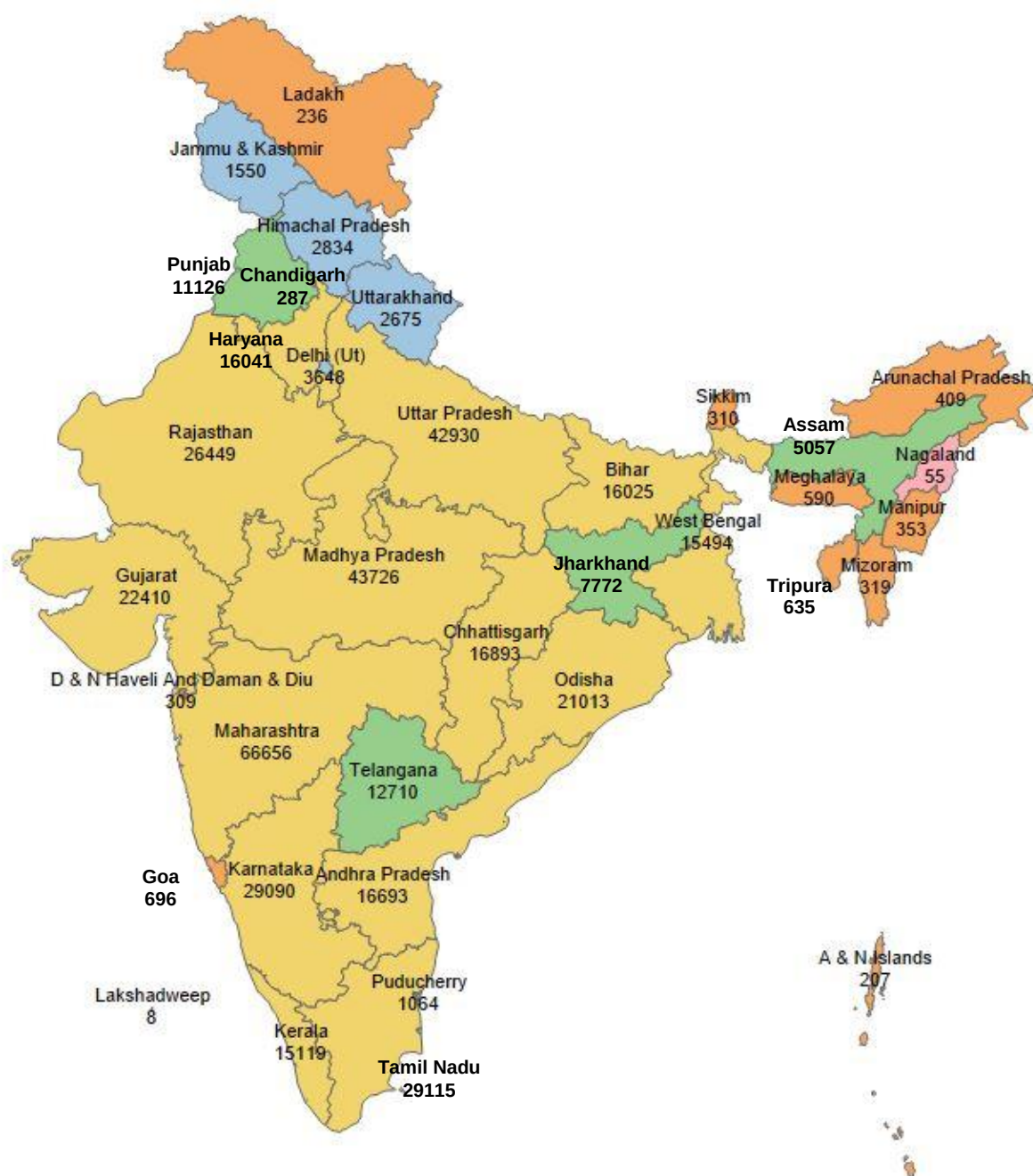
LIST-1.2

Number of Accidental Deaths due to Forces of Nature & Other Causes (2018 to 2022)

Year	Forces of Nature		Other Causes		Total Accidental Deaths	
	Number of Deaths	% change over previous year	Number of Deaths	% change over previous year	Number of Deaths	% change over previous year
(1)	(2)	(3)	(4)	(5)	(6)	(7)
2018	6,891	-3.5	4,04,933	4.0	4,11,824	3.8
2019	8,145	18.2	4,12,959	2.0	4,21,104	2.3
2020	7,405	-9.1	3,66,992	-11.1	3,74,397	-11.1
2021	7,126	-3.8	3,90,404	6.4	3,97,530	6.2
2022	8,060	13.1	4,22,444	8.2	4,30,504	8.3

- As per data provided by States/UTs.

FIGURE-1.2
STATE/UT – WISE ACCIDENTAL DEATHS DURING 2022



- As per data provided by States/UTs.

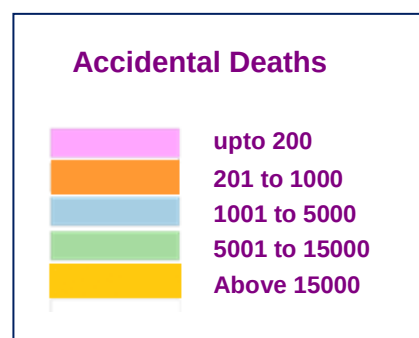
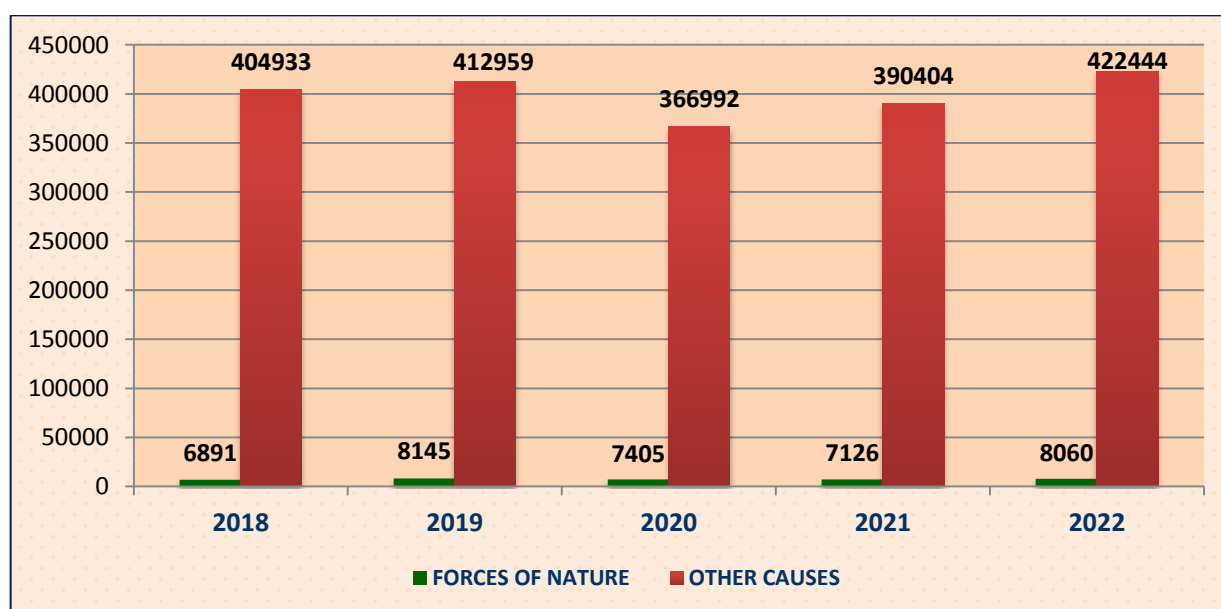


FIGURE-1.3

Total Number of Accidental Deaths due to Forces of Nature and Other Causes during 2018-2022



- As per data provided by States/UTs.

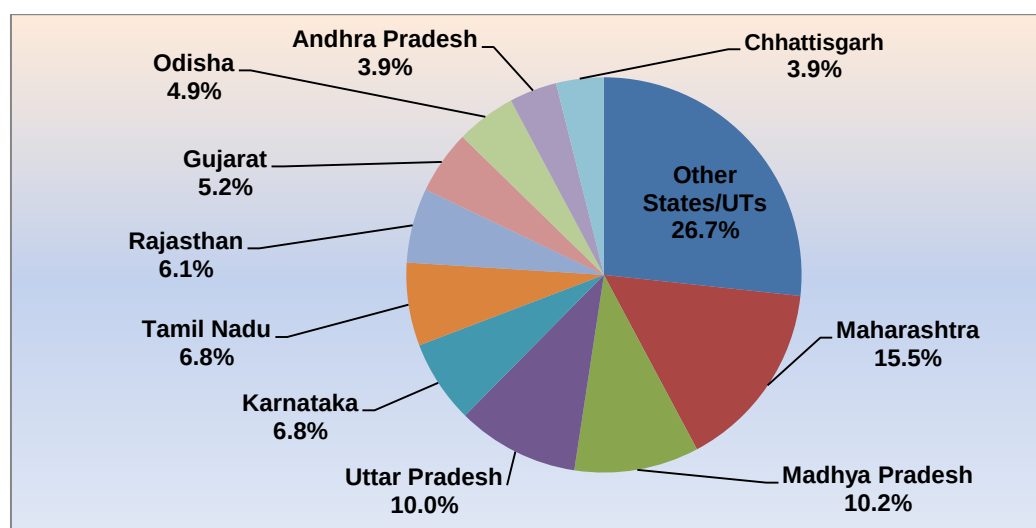
Number & Rate of Accidental Deaths in States/UTs

Maharashtra with a population share of 9.1% has reported highest number of accidental deaths (66,656), contributing nearly one-seventh (15.5%) of total accidental deaths reported in the country. Uttar Pradesh, the most

populous State in the country with population share of 17.0% has accounted for (42,930) 10.0% of the total accidental deaths in the country. The other States having higher percentage share in total accidental deaths were Madhya Pradesh (10.2%), Karnataka (6.8%), Tamil Nadu (6.8%), Rajasthan (6.1%), Gujarat (5.2%), Odisha (4.9%), Andhra Pradesh (3.9%) and Chhattisgarh (3.9%) [Table-1.2].

FIGURE-1.4

Percentage Share of States/UTs in Accidental Deaths during 2022



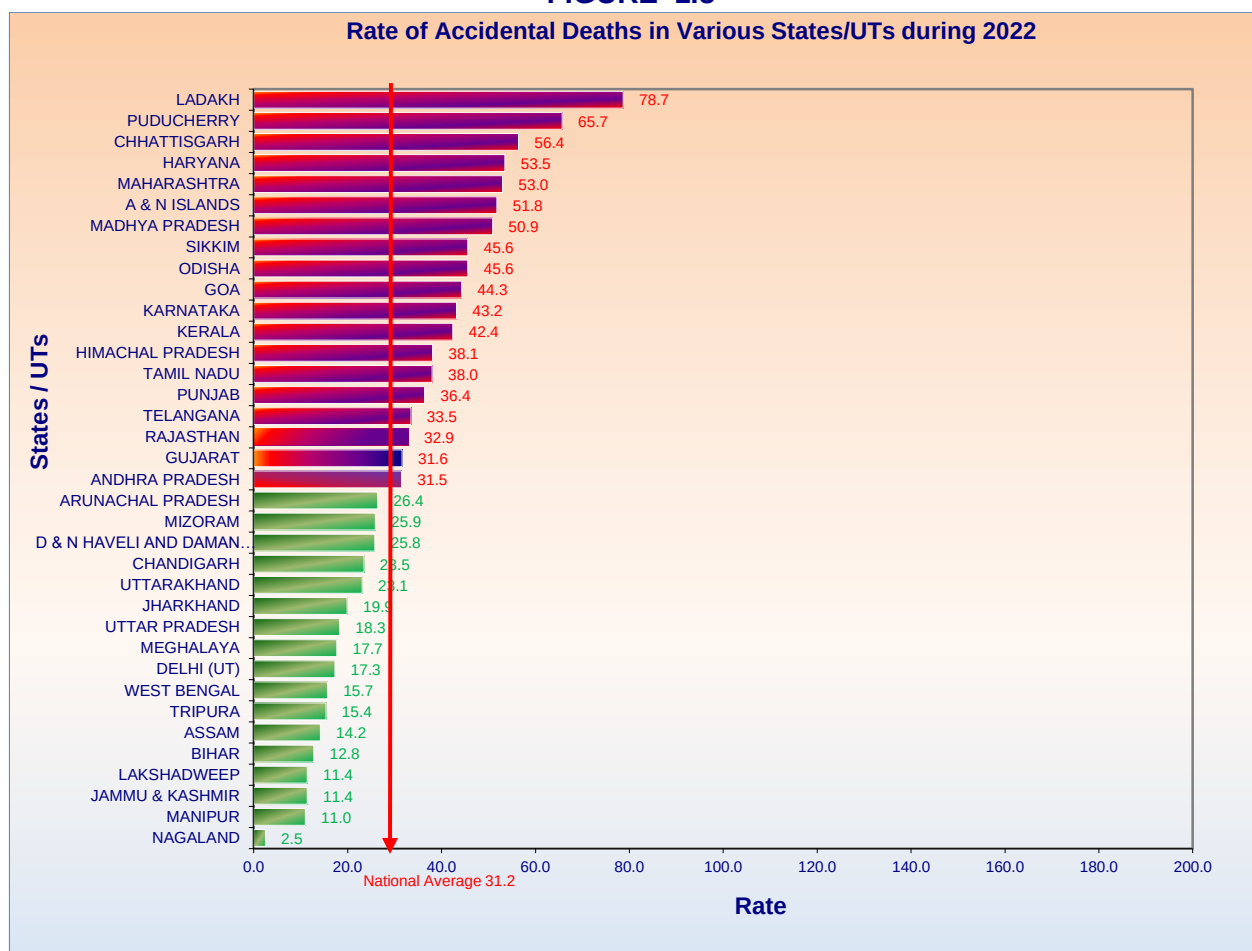
- As per data provided by States/UTs.

Other States/UTs include, Arunachal Pradesh, Assam, Bihar, Goa, Haryana, Himachal Pradesh, Jharkhand, Kerala, Manipur, Mizoram, Meghalaya, Nagaland, Punjab, Sikkim, Telangana, Tripura, Uttarakhand, West Bengal, A & N Island, Chandigarh, D&N Haveli and Daman & Diu, Delhi, Jammu & Kashmir, Ladakh, Lakshadweep and Puducherry

The highest **rate of accidental deaths** was reported from Ladakh (78.7) followed by Puducherry (65.7), Chhattisgarh (56.4), Haryana (53.5), Maharashtra (53.0), A&N Islands (51.8) and Madhya Pradesh (50.9) against the national average of 31.2. Nineteen

(19) out of 36 States/UTs have reported higher rates of 'Accidental Deaths' as compared to All India Average of 31.2 deaths per one lakh of population [**Figure – 1.5**].

FIGURE–1.5



- As per data provided by States/UTs.

Among the States/UTs with significant number of accidental deaths; Bihar has reported the highest percentage increase (48.6%) in total accidental deaths in the year 2022 as compared to 2021 (from 10,781 in 2021 to 16,025 in 2022), followed by Tamil Nadu (18.5%), Uttar Pradesh (15.3%), Jharkhand (13.9%) and Karnataka (10.7%).

Delhi has reported a decline of 40.7% in total accidental deaths in the year 2022 as compared to 2021 followed by Nagaland (35.3%) and Chandigarh (32.0%).[**Table-1.3**]

Gender-wise analysis reveals that male and female constitute 82.6% and 17.4% of total victims respectively. Age group of majority of the victims was between 30 to 45 years. This group of people has accounted for 30.9% of all persons killed in accidents in the country during the year 2022. A total of 52,901 senior citizens (60 years & above of age) have died in various accidents during 2022 comprising 12.3%. [**Table-1.4**].

FIGURE-1.6
STATE/UT – WISE RATE OF ACCIDENTAL DEATHS DURING 2022



Note:

Rate of Accidental Deaths means Number of Accidental Deaths per 1,00,000 population.

- As per data provided by States/UTs.

Accidental Death Rate
All India Average 31.2

	upto 10
	10.1 to 20
	20.1 to 30
	30.1 to 50
	Above 50

Causes of Accidental Deaths

Accidental deaths have been grouped into two broad categories i.e. 'Deaths due to Forces of Nature' and 'Deaths due to Other Causes'. Classification of types and causes of accidental deaths has been done for statistical purpose only.

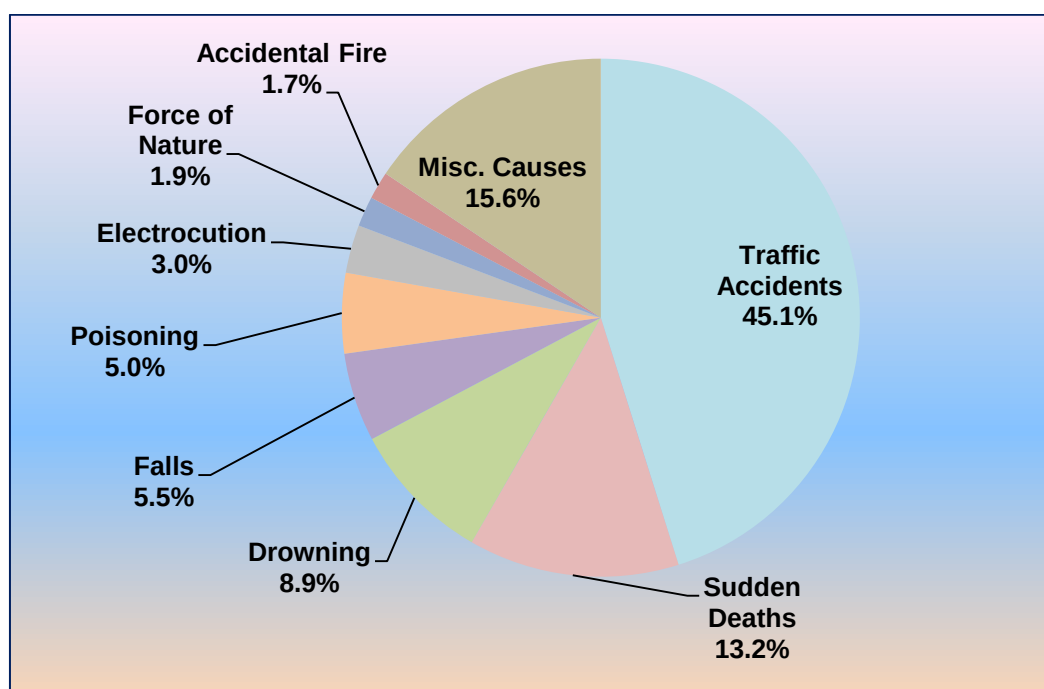
Under causes attributable to 'Forces of Nature', the deaths due to 'Avalanche', 'Exposure to Cold', 'Heat/Sun Stroke', 'Lightning' and 'Torrential Rain' have increased during the year 2022 over the previous year 2021 whereas the deaths due to *inter-alia* 'Cyclone', 'Tornado', 'Tsunami' 'Flood' 'Landslide' and 'Forest fire' have decreased in 2022 over 2021.

The deaths due to 'Air Crash', 'Ship Accidents', 'Collapse of Structure', 'Drowning', 'Electrocution', 'Falls', 'Mines or Quarry Disaster', 'Traffic Accidents', 'Sudden Deaths', 'Deaths of Women during Pregnancy', 'Killed by Animals' and 'Suffocation' have increased whereas the deaths due to Other Causes such as, 'Accidental Explosion', 'Factory/ Machine Accidents', 'Accidental Fire', 'Firearms', 'Stampede', 'Deaths due to Consumption of Illicit/Spurious Liquor', 'Poisoning' and 'Drug overdose' have decreased in 2022 as compared to the previous year 2021.

There has been an increase in number of traffic accidents deaths during 2022 (1,94,347 deaths) over during 2021 (1,73,860 deaths).

Cause-wise details of accidental deaths, its share and percentage variation during 2022 over 2022 are presented in **Table-1.0**.

FIGURE-1.7
Percentage Share of Various Major Causes of Accidental Deaths during 2022
(Forces of Nature and Other Causes)



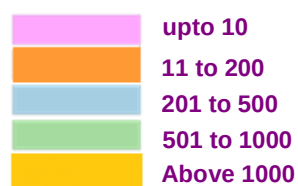
- As per data provided by States/UTs.

Misc. Causes include Collapse of Structure, Suffocation, Deaths of Women during Pregnancy, Killed by Animals, Deaths due to Consumption of Illicit/Spurious Liquor, Accidental Explosion, Drug Overdose, Factory/Machine Accidents, Firearm, Mines or Quarry Disaster, Stampede, Air-Crash, Ship Accidents, Other than above & Unknown Causes.

FIGURE-1.8
STATE/UT – WISE DEATHS DUE TO FORCES OF NATURE DURING 2022



Accidental Deaths



- As per data provided by States/UTs.

Accidental Deaths due to Forces of Nature

Cause - wise accidental deaths due to forces of nature are presented in **Table-1.0**. Out of 8,060 accidental deaths due to forces of nature, 35.8% deaths due to 'Lightning' (2,887 deaths), 9.1% deaths due to 'Heat/Sun Stroke' (730 deaths), 8.9% deaths due to 'Exposure to Cold' (720 deaths) and 6.8% deaths due to 'Flood' (547 deaths) during the year 2022. 'Landslide and 'Torrential Rain' caused 3.3% (269 deaths) and 1.1% (89 deaths) of total deaths due to forces of nature respectively.

The age group-wise distribution of accidental deaths due to forces of nature is given in **Table-1.5**. As per age-wise classification of deaths due to forces of nature, 5.7% of such victims were below 14 years of age (462 out of 8,060 deaths), 4.2% in the age-group of 14 years to below 18 years (341 deaths), 15.1% in the age-group of 18 years to below 30 years (1,214 deaths), 23.7% in the age-group of 30 years to below 45 years (1,908 deaths) and 19.2% victims were in age group 45 years to below 60 years (1,544 deaths) during 2022. Senior citizens (60 yrs. and above) constituted 32.1% (2,591 deaths) of the total such victims.

State/UT - wise and gender - wise distribution of accidental deaths in the country by forces of nature along with their share in total deaths is presented **Table-1.9**. 'Lightning' was the dominant factor and caused 95.7%, 84.7%, 82.6% and 68.6%, of total such deaths in Tamil Nadu (89 out of 93 deaths), Chhattisgarh (210 out of 248), West Bengal (161 out of 195 deaths) and Karnataka (96 out of 140 deaths) respectively. 58.9% (56 out of 95 deaths) of death in Haryana were due to 'Exposure to Cold'. 'Heat/Sun Stroke' was the dominant factor and caused 50.6% (130 out of 257 deaths) of total accidental deaths due to forces of nature in Punjab. 37.7% (23 out of 61 deaths) of deaths in Assam was due to 'Flood'. 'Landslide' was the dominant factor and caused 100.0%, 100.0%, 98.5%, 76.0%, 71.4% and 55.5% of total deaths in Mizoram (6 out of 6 deaths), Nagaland (3 out of 3 deaths), Manipur (64 out of 65 deaths), Arunachal Pradesh (19 out of 25 deaths), Meghalaya (30 out of 42 deaths),

and Himachal Pradesh (61 out of 110 deaths) respectively during 2022.

Majority of deaths due to 'Torrential Rain' were reported in Jammu & Kashmir which accounted for 46.9% (15 out of total 32 deaths) of total such deaths in the country during 2022.

Accidental Deaths due to Other Causes

It is generally accepted that accidents due to 'Other Causes' are preventive and can be reduced by effective safety measures and safety consciousness.

A total of 7,00,688 cases of Accidents due to 'Other Causes' were reported in the country which resulted in 4,22,444 deaths and 4,28,435 persons injured during 2022 [**Table-1.7**].

The cause-wise analysis of deaths revealed that majority of deaths were due to traffic accidents which accounted for 46.0% (1,94,347 out of 4,22,444 total deaths) of total deaths due to 'Other Causes' during the year 2022. Besides, 'Sudden Deaths', 'Drowning', 'Falls' and 'Poisoning' caused 13.4% (56,653 deaths), 9.1% (38,503 deaths), 5.6% (23,786 deaths) and 5.1% (21,606 deaths) of total deaths due to other causes respectively. [**Table-1.0**]

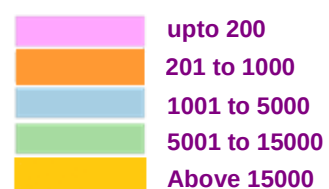
The age-group-wise distribution of accidental deaths in the country by various 'Other Causes' along with their share in total deaths is presented in **Table-1.8**. Majority of victims were in age groups 30 - below 45 years and 18 - below 30 years which accounted for 31.0% (1,30,938 deaths) and 25.1% (1,06,030 deaths) of total such deaths respectively.

State/UT-wise and gender-wise distribution of accidental deaths in the country by various other causes is presented in **Table-1.10**. Maharashtra with 9.1% of country's population has reported 15.7% (66,211 out of 4,22,444 deaths) of accidental deaths due to other causes. Madhya Pradesh, Uttar Pradesh, Tamil Nadu, Karnataka, Rajasthan and Gujarat accounted for 10.1%, 10.0%, 6.9%, 6.9%, 6.2% and 5.3% of total such deaths respectively during the year 2022.

FIGURE-1.9
STATE/UT – WISE ACCIDENTAL DEATHS DUE TO OTHER CAUSES
DURING 2022



Accidental Deaths



- As per data provided by States/UTs.

State/UT-wise and Cause-wise details on 'Accidents due to Other Cause' is presented in **Table-1.10**. State/UT-wise analysis reveals that Uttar Pradesh accounted for maximum deaths (28,615 out of 1,94,347) under traffic accidents. Maharashtra accounted for maximum deaths (19,053 out of 56,653) under 'Sudden Deaths'. Madhya Pradesh accounted for maximum deaths (5,427 out of 38,503) under 'Drowning'. Maharashtra accounted for maximum deaths (4,236 out of 23,786) under 'Falls'.

A total of 507 incidents of consumption of illicit/spurious liquor caused 617 deaths in the country during 2022. States where various such deaths were reported include Bihar (134 deaths) followed by Karnataka (98 deaths), Punjab (90 deaths), Chhattisgarh (60 deaths), Jharkhand (55 deaths) and Uttar Pradesh (50 deaths).

Fire Accidents

Considering the number of fire accidents in the country and potential of the damage that can be caused by such accidents, the Bureau has also collected data on fire accidents according to places of occurrence for appropriate preventive strategies viz. 'Fire in School Buildings', 'Fire in Commercial Buildings', 'Fire in Residential / Dwelling

Buildings', 'Fire in Government Buildings', 'Fire in Mines', 'Fire in Factories', 'Accidental Fire in Trains' and 'Accidental Fire in Vehicles' [**Table-1.11**].

A total of 7,566 cases of fire accidents were reported in the country during 2022, showing a decrease of 10.9% as compared to 2021 (8,491 cases). 7,566 incidents of fire accidents caused injuries to 329 persons and 7,435 deaths during 2022. The place of Occurrence-wise analysis of fire accidents revealed that 53.5% of total deaths (3,979 out of 7,435) were reported in residential/dwelling buildings during 2022. [**Table 1.12**].

Large number of cases of fire accidents were reported in Odisha (1,222 out of 7,566), accounting for 16.2% of total such cases during 2022 [**Table-1.12**].

The State/UT-wise analysis of place of fire accidents revealed that 21 States & UTs have reported 50.0% or more deaths due to 'Fire in residential or dwelling building' during 2022. These States/UTs were Meghalaya, Nagaland, Tripura (100.0% each), Bihar (95.2%), Punjab (92.5%) and Assam (86.5%). [**Table-1.12**].

Accidents in Cities

The term 'Mega City' refers to a city which has a population of 10 lakhs or more. As per the Population Census 2011, the status of mega city has been accorded to 53 cities.

A total of 62,995 accidental deaths were reported in 53 mega cities during 2022. Maximum numbers of accidental deaths (8,286 deaths) were reported in Mumbai accounting for 13.2% of total deaths, followed by Delhi City (5,199), Pune (4,665), Chennai (3,859), Bengaluru (3,732) and Bhopal (3,262).

Rate of accidental deaths in these 53 mega cities was higher (39.2) than all India rate of accidental deaths (31.2). The rate of accidental deaths was highest in Bhopal (173.5), followed by Prayagraj (112.0), Jaipur (102.2), Pune (92.4), Raipur (92.0) and Nasik (88.1) as compared to average rate for the cities (39.2) [Table-1.2].

Out of 62,995 accidental deaths in 53 mega cities; 670 and 62,325 deaths were due to 'Forces of Nature' and 'Other Causes' respectively during 2022. An increase of 25.7% was observed in deaths due to forces of nature and an increase of 17.8% was witnessed in deaths due to 'Other Causes' during 2022 over 2021 respectively. [Table-1.3]

Kolkata, the third largest city in terms of population (8.8% of the total population of all mega cities) among 53 cities, reported the low share of deaths in cities accounting for 1.3% of total accidental deaths in 53 mega cities during 2022. The comparative reporting of 'Accidental Deaths' during 2022 in the top four-metropolitan cities is presented in LIST-1.3.

Accidental Deaths due to Forces of Nature in Cities

A total of 670 deaths due to forces of nature were reported in these 53 mega cities during 2022. Among specified causes of accidents due to forces of nature, maximum deaths were reported due to 'Exposure to cold' followed by 'Heat/Sun stroke', accounting for 22.2% and 19.1% of total such accidental deaths. Due to 'Exposure to cold' maximum deaths were reported in Ludhiana (32 deaths), Amritsar (31 deaths) and Ranchi (26 deaths). Due to 'Heat/Sun stroke' deaths were reported in Amritsar (63 deaths), Ludhiana (22 deaths) and Dhanbad (7 deaths).

Accidental Deaths due to Other Causes in Cities

A total of 1,13,097 cases of 'Other' accidents were reported in 53 mega cities which caused injuries to 57,794 persons and 62,325 deaths during 2022. The cause-wise analysis of deaths due to other causes revealed that 28.4%, 28.1%, and 7.0% deaths in 53 cities were due to "Traffic Accidents" (17,680), Sudden deaths' (17,522), and 'Falls' (4,391) respectively.

Among 53 cities, Mumbai city (8,286) accounted for 13.3% of total deaths reported under 'Other' accidents followed by Delhi city (5,198) and Pune (4,665) which accounted for 8.3% and 7.5% respectively during 2022.

"Data on City-wise information is available on our website under 'Additional Tables for ADSI'"

LIST-1.3

Rate of Accidental Deaths in Metropolitan Cities during 2022

Sl. No.	City	Population (In Lakhs)	Share of Population In Cities (Percent)	Share of Deaths In Cities (Percent)	Rate (No. of Deaths Per One Lakh Population)
(1)	(2)	(3)	(4)	(5)	(6)
1.	Chennai	87.0	5.4	6.1	44.4
2.	Delhi	163.2	10.2	8.3	31.9
3.	Kolkata	141.1	8.8	1.3	5.6
4.	Mumbai	184.1	11.5	13.2	45.0

- As per data provided by States/UTs.
